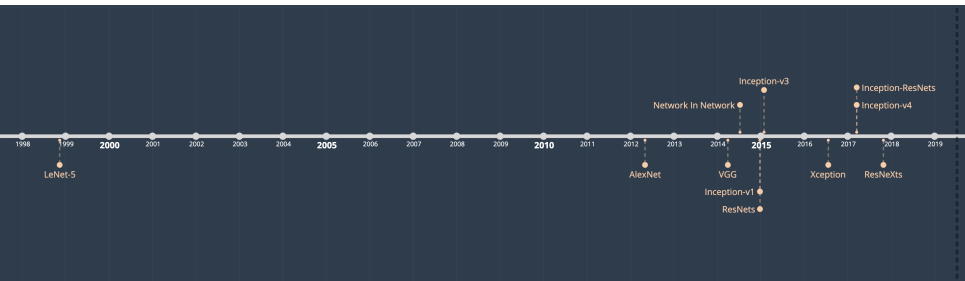


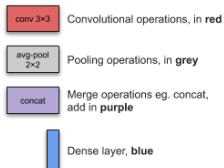
Popular CNN architectures

History



Notations

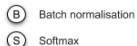
Layers



Activation Functions

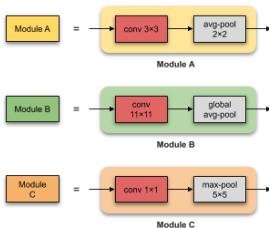


Other Functions

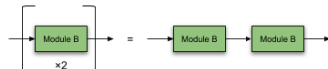


Modules/Blocks

Modules (groups of convolutional, pooling and merge operations), in **yellow, green, or orange**. The operations that make up these modules will also be shown.



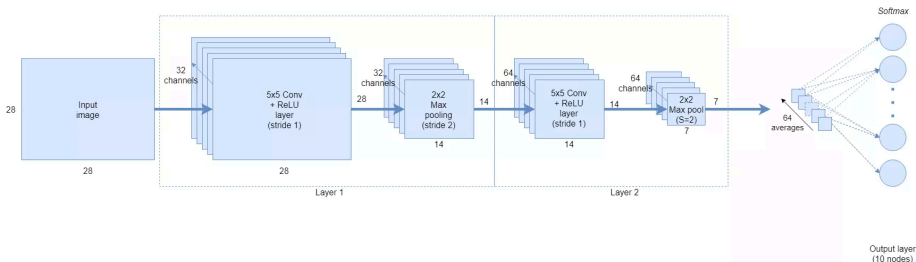
Repeated layers or modules/blocks



Global Average Pooling (GAP)

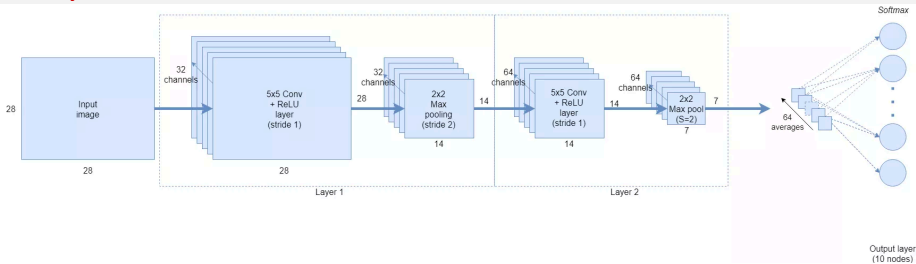
GAP is an operation that calculates the **average output of each feature map** in the previous layer:

- reduces the data significantly and prepares the model for the final classification layer.
- no trainable parameters.



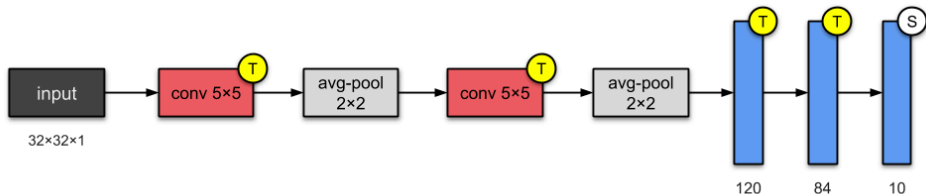
GAP in a CNN architecture

Why GAP?



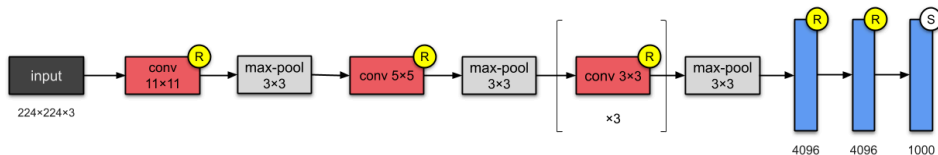
- Less number of trainable parameters: 64 (non-trainable) vs. $7 \times 7 \times 64 \times N$ (FC, trainable) $\rightarrow$ speed up factor.
- Replace FC $\rightarrow$ the feature maps are forced to be more closely related to the classification categories.
- Draw back: lack of complexity compared to FC.

LeNet (1998)



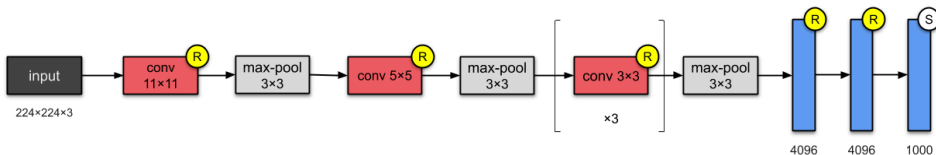
- 2 convs + 3 FCs (hence “5”)
- 60,000 parameters.
- (New) Stacking conv and pooling layers, ending the network with FCs.

AlexNet (2012)



- 5 convs + 3 FCs.
- 60M parameters.
- First to implement Rectified Linear Units (ReLUs).
- Make use of Overlapping Pooling

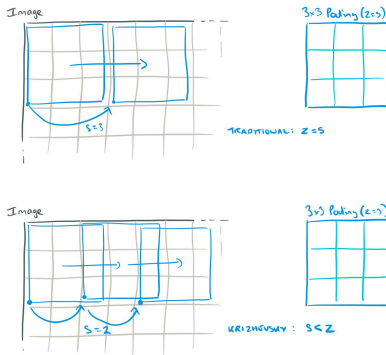
AlexNet (2012) – Overlapping Pooling



3.4 Overlapping Pooling

Pooling layers in CNNs summarize the outputs of neighboring groups of neurons in the same kernel map. Traditionally, the neighborhoods summarized by adjacent pooling units do not overlap (e.g., [17, 11, 4]). To be more precise, a pooling layer can be thought of as consisting of a grid of pooling units spaced s pixels apart, each summarizing a neighborhood of size $z \times z$ centered at the location of the pooling unit. If we set $s = z$, we obtain traditional local pooling as commonly employed in CNNs. If we set $s < z$, we obtain overlapping pooling. This is what we use throughout our network, with $s = 2$ and $z = 3$. This scheme reduces the top-1 and top-5 error rates by 0.4% and 0.3%, respectively, as compared with the non-overlapping scheme $s = 2, z = 2$, which produces output of equivalent dimensions. We generally observe during training that models with overlapping pooling find it slightly more difficult to overfit.

AlexNet (2012) – Overlapping Pooling



- Non-overlapping pooling: spatial information is quickly lost and the network "sees" only the dominant pixel values.
- Overlapping pooling: less loss of surrounding spatial information.

Question: to have overlapping pooling, should we reuse traditional (non-overlapping) pooling with a larger pooling kernel?

AlexNet (2012)

Question: to have overlapping pooling, should we reuse traditional (non-overlapping) pooling with a *larger pooling kernel*?

No, large pooling kernel leads to information destruction.

1D feature

0	0	5	0	0	6	0	0	3	0	0	4	0	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---

$z=2, s=2$

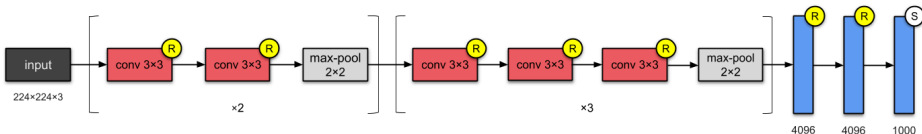
0	5	6	0	3	4	0
---	---	---	---	---	---	---

$z=3, s=2$

5	5	6	3	3	4	0
---	---	---	---	---	---	---

Grid-like pattern disappears!

VGG16 (2014)



- 13 convs + 3 FCs.
- 138M parameters.
- ReLU activation functions as in AlexNet.
- Deeper network (twice as deep as AlexNet).